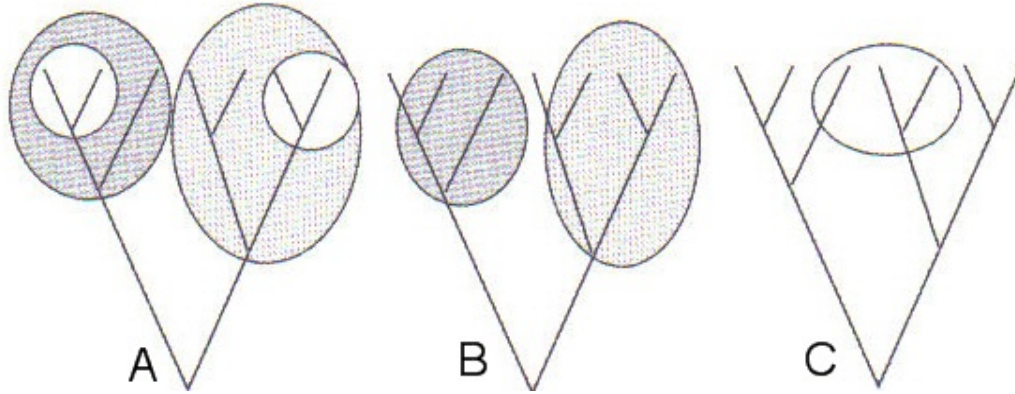


1. Name the three different phyletic groupings that are presented below. Note: Each tree may have more than one “bubble” drawn on it, but each tree (A, B, and C) just has one TYPE of phyletic groupings.



- 1a. _____-phyletic groupings
- 1b. _____-phyletic groupings
- 1c. _____-phyletic grouping
2. Taxonomy, or the classification of organisms based on similarities or differences, dates back to the 1700s. The person who is associated with developing a classification used today was whom?
- _____
3. List the 8 terms below in order from MOST INCLUSIVE (space “A” below; broadest category) to LEAST INCLUSIVE (space “H” below; smallest, most defined category):

class, domain, family, genus, kingdom, order, phylum, species

- | | |
|----------|----------|
| A. _____ | E. _____ |
| B. _____ | F. _____ |
| C. _____ | G. _____ |
| D. _____ | H. _____ |

4. Based on the information in the table below, construct a cladogram (or tree) which depicts the relatedness of the dwarfs based on ancestral or derived characters.

Place each of the dwarfs “G, S, H, D, and B” in the **boxes** on the terminal positions of the tree below this table. Also, place the character state using the character numbers (“1” through “5”, one number will **NOT** be used) **inside the arrows** which point to the dashes that intersect the tree. For each of these numbers, you should also indicate whether the character **APPEARED** or **DISAPPEARED** by placing a plus “+” or minus “-“ after each number. For example, if you thought that **BEARDS APPEARED** for arrow “4f”, you would indicate “**1+**” inside of that arrow. If you thought that **BEARDS DISAPPEARED** for arrow “4f”, you would indicate “**1-**” inside of that arrow.

	Beard (1)	Blue eyes (2)	High IQ (3)	>5 ft. tall (4)	Big nose (5)
Grumpy (G)	no	no	no	no	yes
Sleepy (S)	no	no	no	no	no
Happy (H)	no	yes	no	no	yes
Doc (D) ***	yes	no	no	no	no
Bashful (B)	no	yes	yes	no	yes

*** for cladistic purposes, Doc will be considered the OUTGROUP since he is the OLDEST

